組別:　1　組員1(學號姓名):C108110249 林宇廷　組員2:C108110259 陳汶豪　日期03/30

實驗七：LED shift using HDL

題目: (1) 8 bit led,
(2) 左移次數: 自己學號最後 1 碼 + 2。
換　右移次數: 自己學號最後 1 碼 + 3　or (1 次),　重複 左移, 右移.

一、實驗目的：

1. Using the HDL to design: led shift left and right.

二. HDL code and testbench:

```
module lab07 (clk,  reset, led);
//
input    clk, reset;     // reset signal:    enable: low
output   [7:0]led;       // led output with 8 bits

reg [7:0]led;    // define the led register with 8 bits
  reg [4:0]counter_no;
reg state;
//reg    [23:0]count;   // The count register is for dealy
//
always @(posedge clk or negedge reset)       // at the rising edge
begin
     if(~reset)   // the reset is low
       begin        // inital setting, led=1
          counter_no=0;
          led = 8'h01;
          state=0;
       end
     else


       begin   // most of the time will enter this begin - end
          if(counter_no<2 && state==0)
             begin
          led = led << 1;// shift right one bit of the led rgister
               if(led==8'h00)begin
                    counter_no=counter_no+1;
                    led = 8'h01;
```

```
            end
            if(counter_no==2)
               begin
                  counter_no=0;
                  state=1;
                  led = 8'h80;
               end
         end

      if(counter_no<3&& state==1)
         begin
      led = led >> 1;    // shift right one bit of the led rgister
            if(led==0)begin
            counter_no=counter_no+1;
               led = 8'h80;
            end

            if(counter_no==3)
               begin;
                  counter_no=0;
                  state=0;
                  led = 8'h01;
               end

         end
      end




end

endmodule

=============================================================================/
/ Verilog test

`timescale 1ns / 1ns
```

```
module lab07testbench;

// Inputs
     reg clk;
     reg reset;
     reg state;
   reg [4:0]counter_no;



// Output
     wire [7:0]    led;    // output from lab07
//      wire              counterdiv;

// Call the module of lab 07
//

  lab07    u1(clk, reset, led);

// Initialize Inputs

    initial
clk=0;
always
#2 clk=~clk; // t=2 ns , 500 mhz

  initial
  begin

  #0 reset=0;    // reset the signal
  #20 reset=1;
  #500 $finish;

  end


initial begin

      $dumpfile("dump.vcd");
      $dumpvars(1);
```

end

endmodule

三、EDA Playground simulation Waveform

重複做兩次上數接者三次下數



